

Mohamad Al Abbas

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EDUCATION

Ph.D., Demography and Sociology, University of Pennsylvania Expected May 2027

- Dissertation: *Envelopes of Exposure: Natural, Built, and Socio-Structural Environments as Mediators of Children's Environmental Health*
- Committee: Emily Hannum (co-chair), Jere Behrman (co-chair), and Brian Thiede
- Anticipated defense: Spring 2027

M.A., Urban Spatial Analytics, University of Pennsylvania, Weitzman School of Design Expected May 2027

M.A., Demography, University of Pennsylvania, School of Arts & Sciences Expected May 2027

M.A., International Affairs, Lebanese American University May 2023

B.E., Electrical Engineering, Lebanese American University August 2019

RESEARCH INTERESTS

Environmental demography • spatial demography and spatial epidemiology • child health and human development • environmental justice • spatial methods & GIS • social determinants of health • demographic and quantitative methods

PUBLICATIONS

Peer-Reviewed Journal Articles

AlAbbas, Mohamad (2026). "Housing and malaria risk: Evidence across ecological and climatic gradients in sub-Saharan Africa." *Social Science & Medicine* 403 (2026): 119410.

AlAbbas, Mohamad, Emily Hannum, Jere Behrman, and Leticia Marteleto (2026). "Who is exposed and who is harmed? Social disparities in flood exposure and impact in Pernambuco, Brazil." *Population and Environment* 48.2 (2026): 10.

Manuscripts Under Review

AlAbbas, Mohamad (under review). "Malaria masks the health benefit of community sanitation across 32 countries." *EcoHealth*

AlAbbas, Mohamad (under review). "Capital Without Labor: The Dispossession of the Commons and the Cultural Politics of Consent." *Theory and Society*

Work in Progress

AlAbbas, Mohamad, Emily Hannum, and Jere Behrman (In preparation). "Terrestrial Ecosystems and Climate-Related Child-Health Risks in 37 Low- and Middle-Income Countries." *Demography*

Al Abbas, Mohamad (in preparation). "Racial Demographic Transition, Political Representation, and the Reallocation of Environmental Hazard in U.S. Counties, 1980–2020." *Dissertation Chapter Three - American Journal of Sociology*

AlAbbas, Mohamad, Emily Hannum, and Jere Behrman (In preparation). “The Hydroclimate of the First 1,000 Days: Precipitation and Thermal Anomalies as Distinct Correlates of Early Childhood Development Across 15 Low- and Middle-Income Countries.” *Nature Human Behavior*

AlAbbas, Mohamad, Anirban Dutta, and Andrew Steenhoff (in preparation). “Seasonality and Weather-Anomaly Structure of Severe Pediatric Tuberculosis Hospitalization in the United States, 2005 to 2025.” *The Lancet*

AlAbbas, Mohamad, and Jonathan Stryko (in preparation). “Weather anomalies and crowding as drivers of multidrug-resistant organism colonization in a Botswana neonatal unit.” *The Lancet*

Earlier Peer-Reviewed Publications: Higher Education, Scientometrics, and Policy-Design

Al-Abbas, Mohammad, and Samer S. Saab (2021). “Quantifying a Publisher’s Competence Through Scholarly Engagement.” *Publishing Research Quarterly*

Saab, Samer S., Mohammad Al-Abbas, Roula Diab, Rayana Jaafar, Khaled S. Saab, and Samer S. Saab Jr. (2021). “Setting the Boundaries of COVID-19 Lockdown Relaxation Measures.” *Library Hi Tech*

Al-Abbas, Mohammad, Wissam Fawaz, Raymond Ghajar, Chadi Abou Rjeily, Iyad Ouass, Zahi Nakad, George E. Nasr, and Samer S. Saab (2020). “Toward an Improvement of Engineering Teaming Skills Through an In-House Professionalism Course.” *IEEE Transactions on Education*

Khalife, Joe J., Mohammad Al-Abbas, and Samer S. Saab (2020). “Formation of Policies Guided by Multivariable Control Theory.” *Operations Research Perspectives*

GRANTS AND FUNDED RESEARCH

Principal Investigator, PSC Quartet Award, National Institutes of Health 2026–2027

- Amount: USD 20,000
- Title: Racial Transition, Environmental Burden, and Health: A longitudinal Study of U.S. Counties
- Description: Funds the third dissertation chapter on historical residential segregation, political representation, and environmental exposure disparities using harmonized census tracts and linked full-count census data, 1980–2020.

Researcher, Deocde Zika and COVID, Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) 2024–2026

- Served as a researcher developing environmental exposure data to support re-application for supplementary NIH grants.
- Supported three manuscripts on topics from fertility to migration.

Researcher, PIRE: Climate risk, pollution, and childhood inequalities in low- and middle-income countries, National Science Foundation) 2023–2025

- Served as a research assistant in the development of manuscripts of climate change and child health

TEACHING

Courses Taught

SOCI 2010: Social Statistics, University of Pennsylvania — *instructor of record* Summer 2026

- Designed the course from scratch: lectures, weekly tutorials, simulated datasets, and assessment structure.
- Built the course around deliberately messy data and a single cumulative deliverable, a research paper and proposal assembled one design component at a time.

- Weighted assessment toward interpretation and communication rather than code production, on the assumption that students will use AI tools to write code.

Principles of Monitoring & Evaluation in International Education Development, University of Pennsylvania — *teaching assistant* Fall 2025

- Quantitative course offered by the Graduate School of Education
- Led weekly lab sections on regression-based inference and interpretation; wrote Stata tutorials, reproducible code templates, and annotated walkthroughs for students with varied quantitative preparation.
- Advised applied evaluation projects on research design, variable construction, and the limits of what an estimate supports.

Courses Prepared to Teach

• **Methodological Courses**

- *Basic Demographic Methods* (graduate): demographic rates and person-years, age standardization and decomposition, single and multiple decrement life tables, stationary and stable populations, population projection.
- *Advanced Demographic Methods* (graduate): Leslie matrices, multistate life tables, non-stable variable- r methods, multivariate and functional decomposition, event history and survival models, indirect estimation, sample survey methods, heterogeneity and selection, and standard errors for complex demographic estimators.
- *Social Statistics* (undergraduate, introductory or service): descriptive statistics and data visualization, probability and sampling distributions, estimation and confidence intervals, hypothesis testing, t-tests, ANOVA, chi-square and contingency tables, correlation, and simple and multiple linear regression with attention to assumptions, diagnostics, and interpretation.
- *Regression and Statistical Inference* (advanced undergraduate or graduate): multiple regression, categorical and limited dependent variables, interaction and non-linearity, panel and fixed effects models, multilevel models, and an introduction to causal inference designs including difference-in-differences, event study, and matching.
- *Spatial Methods for Social Scientists* (graduate): spatial data structures, projection and boundary harmonization, spatial autocorrelation and cluster detection, spatial regression, and the linkage of environmental and administrative data to population datasets.
- *Data Science for Social Research* (undergraduate or graduate): reproducible workflows in R and Python, version control, data cleaning and merging, working with administrative and geospatial sources, and prediction versus explanation.

• **Substantive Courses**

- *Population and Society* (undergraduate: fertility, mortality, and migration as the engines of population change; the demographic and epidemiological transitions; population aging and sub-replacement fertility; urbanization; and the recurring public arguments about population size, growth, and decline.
- *Mortality and Population Health* (graduate): cause-of-death structure and the epidemiological transition; the social determinants of mortality; early-life conditions and later-life health; selection, frailty, and heterogeneity; and the data and estimation problems that arise where vital registration is incomplete.
- *Environment, Climate, and Health* (advanced undergraduate or graduate): climate variability, extreme events, pollution, and disaster exposure as determinants of health, growth, and fertility; the built environment and housing as health infrastructure; the measurement of exposure and the difference between being exposed and being harmed; and the evidentiary problems in attributing health outcomes to environmental change.

- *Health Disparities and Environmental Justice* (undergraduate or graduate): the social patterning of health by race, class, and place; fundamental cause theory and stratified exposure and sensitivity; the siting of hazards and the allocation of environmental goods in the United States; political incorporation and representation as mechanisms; and how spatial scale shapes what inequality is visible.
- *Global Health and Development* (undergraduate, service): child health and survival in low- and middle-income countries; water, sanitation, and infectious disease; nutrition and early childhood development; and the major survey data infrastructures, including the DHS and MICS programs, on which this evidence rests.
- *Courses Developed*
 - Society, Environment, and Health**, undergraduate seminar — *designed, full syllabus available* 2026
 - Fifteen-week seminar integrating environmental sociology, social epidemiology, human ecology, and climate science, moving across scales from biological pathways of embodiment to neighborhood processes to global systems of production.
 - Five modules, each anchored by an extended case discussion: Flint, air pollution in the Los Angeles basin, Cancer Alley, Hurricane Maria, and Standing Rock.
 - Assessment built on cumulative engagement rather than examinations, culminating in an original research paper in which students identify a data source, specify variables, and outline an analytic design without being required to execute it.
 - No prerequisites; designed to be accessible to students with no background in sociology or environmental science.

FELLOWSHIPS, HONORS, AND AWARDS

Benjamin Franklin Fellowship , Graduate Group in Demography, University of Pennsylvania	2024–2028
Dean's Fellowship , School of Arts and Sciences, University of Pennsylvania	2023–2024
MEPI Tomorrow's Leaders Graduate Fellowship , U.S. Department of State	2021–2023
Best Presentation Award , ICIET 2022, Session 2, Matsue, Japan	2022
Outstanding Researcher Award , Social Sciences and Management, Lebanese American University	2021

ACADEMIC APPOINTMENTS

Graduate Fellow , Center for Undergraduate Research and Fellowships, University of Pennsylvania	2026–present
Statistical Consultant , Department of Sociology, University of Pennsylvania	2023–2025
• Advise doctoral students across the department on research design, model specification, and estimation at every stage of the degree; build and curate analytic datasets; lead workshops on machine learning methods for social scientists.	
Graduate Resident Advisor , Lauder College House, University of Pennsylvania	2023–present

PRESENTATIONS AND INVITED LECTURES

Guest Lecturer , Introduction to Demography, Population Studies Center, University of Pennsylvania	October 2025
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- Doctoral lecture on fertility dynamics in low- and middle-income countries and the emergence of sub-replacement fertility in high-income countries.

Conference Oral Presenter, American Sociological Association Annual Meeting August 2025

- Stratified and gendered impacts of the 2022 flash floods in Pernambuco, Brazil.

Conference Poster, Population Association of America 2025

- Disparate Social Impacts by Satellite and Self-Report: Extreme Precipitation and Reported Flood Impacts in Pernambuco, Brazil

PREVIOUS EMPLOYMENT

Intern, Investigations Section, Office of Audit and Investigations, United Nations Development Programme Sep 2022–Mar 2023

- Case review, evidence collation, open-source research, and drafting of investigation plans, witness statements, and reports.

Intern, Technology Policy Program, Center for International Policy Sep 2022–Jan 2023

- Designed the program's Social Media Harms tracker covering the MENA, African, Latin American, and Southeast Asian regions; authored policy briefs and maintained a legislative tracker.

Research Assistant (Individual Contractor), United Nations Economic and Social Commission for Western Asia Jan 2022–Aug 2022

- Statistical support, imputation, and data curation for the ESCWA social and demographic database covering 22 countries and 300 indicators; raised database completion from 29 to 83 percent over the contract.

Monitoring and Evaluation Officer, Office of Graduate Studies and Research, Lebanese American University Jun 2019–Dec 2021

- Bibliometric analysis of scholarly journals, faculty and unit performance monitoring, and identification of research collaborations across schools.

Student Research Assistant, Department of Electrical and Computer Engineering, Lebanese American University 2015–2019

METHODS, SOFTWARE, AND LANGUAGES

- **Data and design:** large-scale linkage of DHS, MIS, and MICS survey data to spatial and satellite sources; harmonized U.S. census tract and county panels; panel and multilevel designs; item response theory scale construction.
- **Estimation:** high-dimensional fixed effects and multilevel models (reghdfe, feols, xtmlogit, GEE), event history and survival models, event study and staggered-adoption estimators, inequality decomposition (Gini, Theil), distributed lag models.
- **Spatial:** spatial statistics and GIS, satellite climate products (CHIRPS, CHIRTS-ERA5, ERA5), WWF terrestrial biome and ecoregion frameworks, HAND terrain surfaces, network routing and two-step floating catchment accessibility.
- **Software:** R, Python, Stata, ArcGIS, QGIS, Git, SLURM, $\text{\LaTeX}$.
- **Certifications:** CITI Social and Behavioral Responsible Conduct of Research (2023); CITI Social and Behavioral Research (2023); NIH Protecting Human Research Participants (2018).
- **Languages:** Arabic (native), English (full professional).